

Types of Classification Error

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Accuracy as a Fraction

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Human Classifier Simulation

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Rows Columns and Data Types

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Explain the two error types for Confusion matrices. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Count a fresh evaluation independently

Each pair below is reference then prediction: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows. Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

6. Solve the given case

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

Name the four outcomes.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Positive means sports. A sports book predicted other and a non-sports book predicted sports occur. Name each error and its consequence for the sports shelf.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 13 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Separate recommending an unsuitable book from failing to surface a suitable one.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Build the count matrix

Use all eight worked IDs. Draw a matrix with reference rows and prediction columns, both ordered clear then blocked. Place the IDs in cells before writing the counts.

My reasoning and evidence

13. W2 Explain the two error types

Declare blocked positive. Give TP,TN,FP,FN counts and explain V3 and V4 in plain language. Does positive mean a desirable outcome?

My reasoning and evidence

14. W3 Audit totals and equal accuracy

Give the worked row totals, column totals, total count and accuracy. Compare its errors with [[3,0],[3,2]] using the same axes and eight reference records.

My reasoning and evidence

15. W4 Count a fresh evaluation independently

Each pair below is reference then prediction: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows. Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

My reasoning and evidence

16. W5 Interpret a zero count carefully

A report gives zero false positives but omits the evaluated sample size, axes and class definitions. What information is missing, and why does zero not prove that future false alarms are impossible?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Types of Classification Error

A fictional book-condition checker predicts repair or sound.

Use reference rows and prediction columns in order sound, repair. Repair is positive.

Six reference/prediction pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

Reference rows, prediction columns

	Predicted sound	Predicted repair
Reference sound	2	1
Reference repair	1	2

Read a cell using both its row and column labels.

Types of Classification Error

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare axes

Rows: reference sound,repair. Columns: predicted sound,repair.



2. Fill sound row

Two true negatives and one false positive: [2,1].



3. Fill repair row

One false negative and two true positives: [1,2].



4. Read result

Matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: agreement 4/6, repair recall 2/3.

CHANGE ONE THING

Change one repair/sound prediction to repair/repair after a real model revision.

Explain the changed result using the matching lesson.

Which error causes a sound book to be sent for unnecessary repair review?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Use a confusion table when evaluating recommender category predictions and attach concrete consequences to each error.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use a confusion table when evaluating recommender category predictions and attach concrete consequences to each error.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-condition checker predicts repair or sound.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: agreement 4/6, repair recall 2/3.

1. Declare axes
2. Read result
3. Fill repair row
4. Fill sound row

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Read result Matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: agreement 4/6, repair recall 2/3. Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair. Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

Practice 2: explain the deciding evidence

Changed-case practice: Change one repair/sound prediction to repair/repair after a real model revision. Which conclusion is supported by the supplied facts?

1. Repair row becomes $[0,3]$; agreement becomes 5/6.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Repair row becomes [0,3]; agreement becomes 5/6. One false negative moves to true positive; the sound row stays [2,1].

Repair a missing building block

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?. Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated input format, selected rule and recorded output. Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
